THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 30°	Angle of reflection equals angle of incidence
2	**(a)** straight lines in a uniform medium	Rectilinear propagation
3	**(a)** the same plane	Second law
4	**(b)** left appears as right	Seen in all mirrors
5	**(c)** diverge	Diverging mirror
6	**(d)** concave	Magnified, erect image when close
7	**(c)** curved surface(s)	Bends light
8	**(c)** erect and diminished	Smaller view
9	**(d)** concave lens	Diverging lens for myopia
10	**(b)** an erect, magnified image	Object within focus

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle of incidence equals angle of reflection (1). Incident ray, reflected ray and normal lie in the same plane (1).	1+1
12	Line perpendicular to the surface at the point of incidence (1). Angles are measured from it (1).	1+1
13	35° (1). 70° (1).	1+1
14	Erect and same size (1). Laterally inverted and virtual (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle of reflection 25° (1). Reflected ray on opposite side of normal (1). Angle with mirror 65° (1).	1+1+1
16	Place a mirror upright on paper and draw normal (1). Send a ray with a pin or torch and mark the reflected ray (1). Measure angles; they are equal (1).	1+1+1
17	Angle of incidence is 0° (1). Angle of reflection is 0° (1). Ray reflects back along the same path (1).	1+1+1
	OR: Plane: erect, same size (1). Concave: depends on distance (1). Convex: erect, diminished (1).	1+1+1
18	Close: erect and enlarged (1). Far: inverted and smaller or larger (1). In between size changes continuously (1).	1+1+1
	OR: Convex gives wide view (1½). Concave gives enlarged image (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Law 1 (1). Law 2 (1). Set-up with mirror and pins (1). Measuring angles (1). Conclusion (1).	1+1+1+1+1
	OR: Small distance: erect and enlarged (1½). Large distance: inverted (1½). Shaving mirror (1). Solar cooker (1).	1½+1½+1+1
20	Always erect (1). Always diminished (1). Wide field of view (1). Rear-view mirror (1). Shop security mirror (1).	1+1+1+1+1
	OR: Shape (1). Image type (1). Size (1). Lateral inversion (1). Use (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 10: **Light: Mirrors and Lenses** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 55°	Angle of reflection equals angle of incidence
2	**(d)** reflection	Mirrors reflect light
3	**(a)** 0°	Retraces its path
4	**(c)** concave mirror	Like inside of a spoon
5	**(a)** erect and diminished	Smaller view
6	**(c)** erect and enlarged	Shaving mirror
7	**(a)** centre	Bulges outward
8	**(c)** convex lens	Enlarges when close
9	**(c)** convex lens	Converging; real and inverted possible
10	**(c)** convex mirror	Back of spoon

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Concave curves inwards and converges light (1). Convex bulges outward and diverges light (1).	1+1
12	Concave: dentist's mirror or solar cooker (1). Convex: rear-view mirror (1).	1+1
13	They give a wide field of view (1). Image is erect though diminished (1).	1+1
14	Left side of object appears as right in image (1). Seen in all mirrors (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Concave reflectors direct sunlight to a point (1). Light converges onto a cooking pot (1). Heat is produced to cook food (1).	1+1+1
16	Close: erect and enlarged (1½). Far: inverted (1½).	1½+1½
17	Convex enlarges text when close (1½). Concave always makes text appear smaller (1½).	1½+1½
	OR: Keep a pencil in an empty tumbler and see it straight (1). Fill half with water (1). Pencil appears bent due to refraction (1).	1+1+1
18	Magnifying glass (1). Spectacles (1). Cameras, microscopes, telescopes (1).	1+1+1
	OR: Reflection: light bounces back; refraction: light passes with bending (1). Same medium vs different medium (1). Mirror vs lens (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Convex converges (1). Concave diverges (1). Convex images enlarged/diminished (1). Concave images erect, diminished (1). Uses magnifier and spectacles (1).	1+1+1+1+1
	OR: Mirror line (1). Normal (1). Incident ray (1). Reflected ray (1). Angle labels with i = r (1).	1+1+1+1+1
20	Plane gives same size grid (1½). Concave gives enlarged or inverted image (1½). Convex gives smaller grid with wide view (2).	1½+1½+2
	OR: Far away: inverted image (1). Image changes size as she moves closer (1). Close enough: erect and magnified (1). Distance decides image (1). Concave mirror converges (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.